UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 4: Programming Fundamentals (CS-308)

Basic Information			
Lab #:	4	**Teacher:**	Arsal Mahmood
Objectives	1. While Loop 2. Do-while Loop		
Registration#		**Student Name**	
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : While Loop

1. Calculate the sum of numbers from 1 to 100 and display it on screen.
2. Calculate sum of even and off numbers separately from a user given range. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and evens in between 50 and 80 and display on screen.
3. Input the marks of a class of ten students. The marks must be between **0** to **100**. Also check marks should not be less than 0 or greater than 100. Calculate the grades of every student and display it. Also calculate the total of the grades and the class average.
   - >90 = A
   - <90 and >80 = B
   - <80 and >70 = C
   - < 70 and > 60 = D
   - <60 = F
4. Input a number from user and display a pattern like these on screen. For example, user has entered 5, the program should display following pattern on screen.:

   (a)
   ```
   *   *   *   *   *
   *   *   *   *
   *   *   *
   *   *
   *
   ```

P-T-O

(b)

1	2	3	4	5
1	2	3	4	
1	2	3		
1	2			
1				

(c)

1				
1	2			
1	2	3		
1	2	3	4	
1	2	3	4	5

(d)

?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?

(Solve all these questions separately. Also, the pattern would change if the user given number is changed)

Activity 2 : Do-While loop

5. Write a program which inputs from user a character between **a** to **z** and guess whether the character matches with a pre-defined letter or not. If it matches, then display a message "**Congratulations**". It gives the user five chances or tries to guess the character.

6. Calculate sum of even and off numbers by using do-while loop separately from a user given range. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and even in between 50 and 80 and display on screen.
7. Print the table of user given number from 1 upto user given limit. Like if user enters 3, and limit is 4, program should print table of 3 as follows:

3	*	1	=	03
3	*	2	=	06
3	*	3	=	09
3	*	4	=	12

8. Ask the user to input numbers repeatedly until they enter -1.
   - Keep track of the largest number entered.
   - Print the largest number at the end.

   Explanation of code:

   while loop continues until -1 is entered.

   Inside loop, use if condition to compare each number with current maximum.

{end of Lab}